Prime Degree Frontier

Native challenge and admission controls · 15 October 2026

Scope and frozen identity

The challenge ranks the largest independently verified odd prime degree separately for every exact registered standard prime-field source curve. Expanded normalized rational maps and monic kernel polynomials are required. Each leaderboard build runs fresh independent replay; saved scores never enter ranking.

Source revision	9bb7cbce2b84151eff6846b02da842a8ef019343
Rules/catalogue season	19f4503758a3dc2d9764f2283b4c92c7a35651e75ee7157f546b2563fcbf5c30
Executable SHA256	dd84d9213c76eb65aba1cb564769dcb0b06509dba751c8daaf1ad58edcd656f1
Run host	Apple M4 Pro; macOS 26.6; shared host
Run scope	Correctness/admission controls on a shared host; no throughput comparison.

Frozen judge source · Native audit · Raw packet

Observed coverage and controls

Registered standard prime models	54
Construction adapter available	47
Independent replay available	47
Replay adapter unavailable, retained in roster	7
Candidate curve-degree pairs above 10453 through 65537	157
Altered-artifact controls rejected	9
Verified source rows in control leaderboard	2
Short-map scheduling controls accepted / executed	20 / 20
Short-map scheduling controls censored	0

Counts are exact native auditor outputs, not sampled estimates. The low-order Frobenius screen produces candidates only. Missing adapters remain explicit. Two known-answer fixtures score P192/13 and P521/7 on different source rows. These controls contribute zero new discoveries. A forged saved score is rejected.

Exact source: icv1-fp192-t31607402316713927207482677199-52e4af59 verified degree 13.

Exact source: icv1-fp521-t65787750189432823735744433231502011753692325721938727626347220121939840805170 3-f767e0d1 verified degree 7.

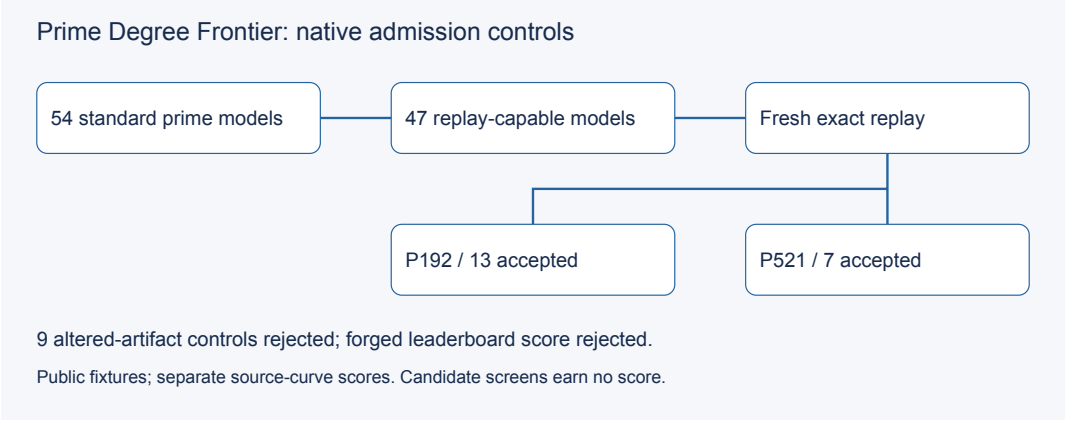


Figure 1: Native admission-control flow. Roster and control counts come from audit.json.

Rejection evidence and resource policy

Control	Observed result
composite	REJECTED
inflated-degree	REJECTED
source	REJECTED
kernel	REJECTED
map	REJECTED
compact-only	REJECTED
traversal	REJECTED
missing-hash	REJECTED
duplicate-kernel	REJECTED

Kernel, numerator and source mutations are tested after recomputing their receipt hashes. Other controls alter the degree, substitute a compact-only claim, traverse a curve path, or omit a hash. A prewritten PASS and an inflated stored score do not change the freshly verified degree. Output-directory reuse is refused and the first judgement's bytes remain unchanged.

The ninth case repeats one valid kernel in both map slots. COMPLETE coverage requires distinct normalized kernels and distinct Frobenius roots; that false completeness claim is rejected. Partial maps remain eligible.

Replay has a 3600-second wall budget and an 8 GiB sampled RSS cap per submission. A lower replay timeout is permitted. The bundled constructor has a separate 1800-second/8 GiB limit. The raw map limit is 256 MiB, summary 1 MiB and stderr 8 MiB. Resource failures are censored with no score; invalid maps are rejected. Snapshot copying precedes the bounded child process. Operational receipts are retained; no isolated timing or peak-memory guarantee is inferred.

The bundled constructor accepts degrees through 1000000. The judge accepts u64 prime degrees only when the full explicit artifact fits the byte and verification limits. These implementation limits do not prove nonexistence. External constructor costs are not independently attested or used as a tiebreak.

Fresh starting leaderboard

These historical maps were replayed again under this run's pinned season. Each degree scores only its own exact source. The short-budget P192/10453 control returned CENSORED with no score; the full-budget leaderboard used a separate fresh replay.

Exact source ICV1	Prime degree	Coverage	Maps
icv1-fp192-t31607402316713927207482677199-52e4af59	10453	PARTIAL	1
icv1-fp224-t473310010854560191642182 7343930821-fec01e99	1471	PARTIAL	1
icv1-fp521-t657877501894328237357444 33231502011753692325721938727626347220121939840805170 3-f767e0d1	7	COMPLETE	2

Every admitted map passed the independent kernel, codomain and exact-map checks plus twenty public scalar transports. Re-admitting an existing map contributes zero additional discoveries. The other roster rows stay unscored. Raw packet and fresh per-entry certificates

Degree 10453: icv1-fp192-t31607402316713927207482677199-52e4af59
→ icv1-fp192-t31607402316713927207482677199-e2f29854
Degree 1471: icv1-fp224-t473310010854560191642182734393082 1-fec01e99
→ icv1-fp224-t473310010854560191642182734393082 1-5a35f6f7
Degree 7: icv1-fp521-t657877501894328237357444332315020117536923257219387276263472201219398408051703-f767e0d1
→ icv1-fp521-t657877501894328237357444332315020117536923257219387276263472201219398408051703-5014bab6
→ icv1-fp521-t657877501894328237357444332315020117536923257219387276263472201219398408051703-e804740d

Reproduction and evidence

Use the dedicated ReleaseMe `isogeny` executable. The runtime embeds its catalogue, rules, constructor and replay engines and requires no Cargo checkout. Publication of this new command remains subject to the main release workflow.

```
isogeny challenge policy
isogeny challenge plan --curve all --from 10454 --to 65537
isogeny challenge run --curve p192 --ell 13 --out fresh-control
isogeny challenge judge --input submission --out fresh-judge
isogeny challenge leaderboard --input submissions.json --out fresh-board
```

Source-linked evidence lives beside this report: audit.json, freeze.json, compressed policy/plan JSON and the complete raw archive. That archive contains the native control packets, historical-anchor replay and attempt receipts. The Markdown report identifies exact archive paths and test outcomes. audit.rs reproduces coverage and rejection counts.

Limits and related graphics

This is a challenge for pinned registered models, not a worldwide-record claim or a finite mathematical maximum. Full explicit maps are its sole accepted representation. Compact or custom-curve tracks require a separate contract. The IC/rho scoreboard, IC leaderboard, progress timeline and canonical curve edges remain unchanged: this packet establishes admission behavior and retains known-answer maps without a new IC ratio, new registry model or new curve edge. The earlier CPU optimization and additional-map campaigns remain separate.

The first full-suite attempt censored a valid short replay with zero RSS samples. A later installed run on that source passed, confirming a scheduling-dependent startup race. The fixed source makes the replay child wait until its supervisor records positive RSS. The follow-up packet adds twenty fresh judgements of the same public P192/13 map, each requiring a real sample. Earlier failed build/test logs and the pre-fix packet remain archived; they are not rewritten as passes. The intermediate stress packet accepted 19 of 20 and censored one during OS exit cleanup. The final correction uses a completion marker after all algebra and failure handling. Any missing-monitor refusal during active work stays censored; a later passing certificate never erases it. Zero executed stress controls in the table means that the earlier run did not request this phase.